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$$\begin{aligned}\sum_{n \geq 2}^{\infty} \frac{1}{(\ln n)^{\ln n}} &= \sum_{n \geq 2}^{\infty} \frac{1}{e^{\ln n \ln(\ln n)}} = \sum_{n \geq 2}^{\infty} \frac{1}{n^{\ln(\ln n)}} \\ \Rightarrow \sum_{n \geq 2}^{\infty} \frac{1}{n^{\ln(\ln n)}} &\leq \sum_{n \geq 2}^{\infty} \frac{1}{n^2} \\ \sum_{n \geq 2}^{\infty} \frac{1}{n^2} \text{ convergent} &\Rightarrow \sum_{n \geq 2}^{\infty} \frac{1}{n^{\ln(\ln n)}} \text{ ook.}\end{aligned}$$

References